

HEARTLAND VIDEO EXCERPT TRANSCRIPT

- 1865. The Great Plains. Where 30 million buffalo roam.
 - Vast, untouched, a wilderness dividing America.
 - Crossing the continent takes six months. 20,000 die on wagon trains.
 - By ship, it's an 18,000-mile journey around South America.
 - To conquer the wilderness and unite East and West, President Lincoln green-lights a **transcontinental railroad**, 2,000 miles long.
 - It will transform the nation triggering a tidal wave of settlement across the Great Plains.
 - America's ancient wilderness meets modern American steel and muscle.
 - An army of hammer-wielding men, Irish immigrants.
 - Civil War vets. Railway men. Their mission: to tame nature itself.
 - The biggest obstacle heading east from California a 12,000-foot wall of granite, the Sierra Nevada.
 - With the Pacific of North America plates collide, billions of tons of ancient rock rise up crumpled like tinfoil.
 - Over the last 4 million years, the Sierra Nevada mountains climb more than 2 miles high.
 - They're still growing. 13 Feet in 1,000 years. One day, they could rival the Himalayas.
 - Only a mad man could dream of running a railroad across mountains like this.
 - They don't call him crazy Judah, for nothing. Obsessed with the railroad, he sees a way through.
 - "Come on down, boys." Theodore Judah makes 23 trips into the peaks. And peg that, boys.
 - Plotting a path across ridges and through mountain summits, building it will be the engineering challenge of the century. "Yup, let's mark that." "This is the most magnificent project ever conceived. An enterprise more important to the people of the United States than any other. The railroad will be built, and I will have something to do with it."
 - With Judah's route approved, two companies begin work.
 - The Union Pacific starts from Omaha in the east; the Central Pacific from Sacramento in the west.
 - They'll meet in Utah.
 - It'll cost over \$2 billion in modern money, but the government doesn't have enough cash.
 - It pays the companies in federal land.
 - They must finish in 15 years or lose everything.
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- Paid by the mile, adding curve adds profit.
 - Corrupt investors built the railroad for every cent they can, a nine-mile curve means an extra 120 acres of federal land, and they'll end up owning an area the size of Texas.
 - First they must conquer the Donner Pass. 7,500 feet up, the highest Judah's route.
 - Cursed by 30 feet of snow each winter. Avalanches, tragedy.
 - Here just 20 years earlier, the Donner Party became trapped in the snow and ate each other.
 - Now Judah's railroad cuts right through the mountain, 1,649 feet of rock must be excavated, the longest tunnel on the route.

- **Chinese** laborers dig day and night.
 - It's easier to ship workers from China than to get Americans across the continent.
 - The transcontinental railroad was built by Chinese workers brought over specifically to work on the railroad, and they were considered somewhere in between human and animal.
 - They were not expected to survive.
 - They were expected to come here and work and die.
 - 7,000 miles from home, 17-year-old Hung Leiwo swaps a life in poverty in Canton for a back-breaking work on the railroad gang.
 - Hung Leiwo must cut through granite so tough, a rock the size of a big toe will support 50 tons of locomotive.
 - Progress slows to inches a day. To breakthrough, they need nitroglycerin, but transporting it is banned when 15 men are blown to pieces.
 - In a mobile lab, Scottish chemist James Houghton mixes it on the spot.
 - Nitroglycerin is 13 times more powerful than gun powder.
 - So unstable...any physical shock, and it will explode in his hand.
 - Houghton gets hazard pay. \$4,000 a month in modern money.
 - After three months in the mountains, he turns to drink, leaving the nitro to Chinese men like Hung Leiwo, Irish crews won't touch it.
 - Detonation creates temperatures of 9,000 degrees. It's hot as the surface of the sun.
 - An estimated 1,500 Chinese die in explosions and rock slides. Hung Leiwo survives.
 - His son will be the first Chinese-American to graduate an engineering from the University of California at Berkeley.
 - Once through the mountains, track laying accelerates from 10 inches to 6 miles a day.
 - Each spike is struck 3 times. 10 spikes to a rail. 400 rails a mile.
 - 21 Million hammer swings complete the railroad.
 - May 10th, 1869. A one-word message arrives by telegraph: "Done."
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- The Internet of the area, the transcontinental railroad, changes everything it touches, triggering a mass migration to the **Great Plains**.
 - The Great Plains is conquered by steel and steam.
 - The transcontinental railroad threads a thin line of civilization through the wilderness. People follow.
 - In just one year, 40,000 settlers move to Nebraska, fanning out across the frontier in wagon trains.
 - When in the mid-19th century to the late 19th century, they went out and settled some hostile territory known to mankind.
 - The Great Plain where I grew up. These were true pioneers.
 - The government accelerates the process with the greatest land giveaway in history.
 - Anyone with a \$10 filing fee can claim free land.
 - A quarter are single women and ex-slaves.
 - When you see the desperate scramble in these rickety wagon trains, you realize that promise of America was land.

- These are people who never in a million years would be able to own land in New York.
 - Eventually 10% of the United States will be given away under the **Homestead Act**.
 - “I’m not going back to Indiana to rent until I bust entirely and have to walk back.”
 - Uriah Oblinger, Civil War vet claims his 160 acres in Nebraska. There’s a catch.
 - 110 degree summers spark prairie fires. Trees can’t survive the drought on the plains.
 - There’s so little rain, nothing grows here but grass.
 - Without lumber to build houses, the new inhabitants live in mud huts.
 - Built of sod, cut from the plains. Uriah dismantles his wagon to make doors and windows.
 - They often had insects. They invited snakes.
 - It was pretty much like living in a burrow in the ground. I think the pioneers did have it hard.
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- “I expect you think we live miserable because we are in a sod house.
 - But I’ll tell you, in solid earnest, I never enjoyed myself better.
 - Every liquid we strike is for ourselves and not half of someone else’s.
 - The devout Oblingers face daily tests of their faith.”
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- With no mountains to stop the wind, the Great Plains are a breeding ground for massive thunder storms. “The most objection I have to the weather here is the wind.
 - There’s a great deal of it during the winter and spring, and being nothing to break it, one feels it more. “
 - “Not too far, honey.” The Oblingers live in tornado alley.
 - More twisters in this region than anywhere else on earth.
 - “Honey!” Over 400 touch down every year “Tell the folks they never seen a storm in Indiana, only playthings.”
 - 200 miles an hour winds spin into a vortex, sucking in air and anything not bolted down.
 - In 1930, a man is carried a mile across the Kansas, fish and toads rain from the sky.
 - The Oblingers hunker down in their heavy sod house, clinging to their new found independence.
 - A prehistoric species emerges to battle for the Great Plains.
 - Locusts. After devouring the local vegetation, they released pheromones that signal it’s time to move on.
 - They grow long wings. Swarms head east on the wind.
 - They join up over the Great Plains and become a plague.
 - In 1874, they devour half of the crops in the West. Three trillion locusts.
 - Half a mile high, 100 miles wide. 1,000 miles long, as big as Colorado. They block out the sun.
 - Agricultural Armageddon. To men like Uriah, the locust are the wrath of God.
 - By 1892, half the population of western Nebraska goes east. Uriah stays.
 - Within 30 years, the locust is extinct.
 - It’s breeding grounds in the Rockies plowed over by settlers like Uriah.
 - In 10 years, the Great Plains become the bread basket of the country.
 - For the first time, America can feed itself.

- Today 50 million tons of wheat is farmed each year, but trees are still scarce, and to build towns settlers need wood.
- In Michigan, Wisconsin, and Minnesota, loggers harvest over 50 million acres of trees.
- Green gold. A magnet for Scandinavian woodsmen.
- Between 1825 and 1925, a third of Norway's entire population comes to America...including Nils Haugen.
- "The pay was \$3 a day. You had to have a good pair of driving boots well caulked to be able to keep on top of the logs."
- There's millions of dollars at stake. If the flow of logs stops, towns can't be built.
- Log jams. River man's ruin.
- In 1886, pines to build 20,000 homes get stopped on the St. Croix River.
- 150 Million feet of wood.
- Remove the right log, and the rest will explode downstream. River-men die clearing the obstructions like this.
- In 1892, 2 billion feet of **lumber** will be cut in Wisconsin alone.
- The railroad feeds lumber into the West's construction boom.
- Towns are built so fast, there's no time to name streets. They're given letters and numbers.
- The Great Plains is also home to the most numerous species of large wild mammal on earth.
- 30 million **buffalo**, herds up to 25 miles long, race to summer breeding grounds.
- On a collision course with the modern world.
- The railroad brings a new kind of hunter to the Great Plains. Driven by profit, fresh from the carnage of the Civil War.
- Two million rifles are in circulation. Over a million veterans trained to use them have a new target in their sights.
- Frank Mayer, Civil War vet. Buffalo hunter. "I had nothing to look forward to in civilization. I was crazy about guns."
- Mayer tracks 2,000-pound buffalo, easily capable of crushing a man. He picks them off from 200 yards.
- "If you could kill them, what they brought was yours. They were walking gold pieces."
- Hunters harvest the buffalo for its hide.
- In 1872, they ship over one million out of Kansas alone, worth \$3 a piece back East.
- On a good day, Mayer earns more than the President.
- Factories use long strong strips of buffalo leather as drag belts. Small pieces become coats and shoes.
- To meet demand, hunters kill 8,000 buffalo a day for their hides alone.
- The Plain Indians depend on the buffalo and worship them.
- "The buffalo were our strength from whence we came, and whose breast we suck as babies all our lives."
- Black Elk is 6 years old when the railroad arrives.
- Unlike the white hunters, his people waste none of their kill. Sinews become bow strings. Bones are cups and spoons. Skin is clothing, tepees and coffins.
- Native Americans and buffalos have co-existed since the last ice age.

- Black Elk's ancestors hunted them on foot. There were no **horses** to ride. The modern horse isn't native to North America.
- Spanish conquistadors brought them from Europe in 1493. Some escaped to the Great Plains.
- Perfect horse habitat. 400 years later, over a million mustangs run wild.
- Taming horses transformed the life of the Plains Indian. They become expert horsemen. "The battle cry went up. Hokahey! Which means "to charge," and the hunters went in for the kill."
- On horse back, the bow is the weapon of choice.
- And the time it takes to reload a gun, a warrior can ride 300 yards, and fire 20 arrows.
- Buffalo can run at 35 miles an hour. Hunts cover hundreds of miles over many days. It can take 15 arrows to kill a buffalo.
- White hunters like Frank Mayer use a single cartridge. He aims for the lungs.
- A clean kill drops a buffalo without disturbing the herd. 30 million are killed in a little over a decade.
- After hunters take the hides, train loads of men arrive to pick their carcasses.
- They make buttons from bones and grind down skeletons for fertilizer and porcelain.
- The primary resource keeping Native Americans alive is gone.
- Facing starvation, they're forced on to reservations.
- In 1889, just 85 wild buffalo exist in the whole United States.
- The men who ride the great iron horse are taming the wilderness.
- The railroad will bring another modern American icon to the Great Plains.